Demo Client · DEMO-0011
13 November 2026 · v1.0

Field	Value
Document	Methodology of the business analytics review
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1. Purpose

To describe, step by step and with every parameter, how each figure of the review is produced, so that someone else can repeat it on the same data and obtain the same numbers.

2. Inputs

Data set	Size	Fingerprint (SHA-256)
purchases	4,320 rows	`1c793efc4eb8d9ee`
stock	1,080 rows	`f399ab5b861a9c05`
ads	2,190 rows	`2141427621ddf485`
branches	3 rows	`7213bda7ecaa7296`
loyalty	2,730 rows	`5cdd91850ba23aa6`
payroll	72 rows	`9ce7401c654058f1`
menu	15 rows	`d9013d82a0d97fe6`
budget	72 rows	`92fb307eeb3613a0`
pos	257,034 rows	`7d843af5e446cb27`
ledger	480 rows	`5366f65dc6e5365a`

Only the transactions are required. Each input is fingerprinted with SHA-256 on arrival and recorded in the run audit trail.

3. From the client's files to standard tables

1. Each source is read as delivered: CSV, Excel (title rows above the header are detected), JSON, Parquet, a database query, an API or a Google Sheet.
2. A written mapping names, for every standard field, the client's column, a constant, or an arithmetic expression of columns. A first mapping is drafted by `scripts/suggest_mapping.py` and corrected by the analyst.

3. Values are converted to their type: currency symbols removed; the decimal separator inferred from each value; Arabic-Indic digits converted; brackets and trailing minus read as negative; Excel serial dates converted; day-first dates detected.
4. A row without a required field is left out; a row matching an exclusion rule is left out; every such row is counted in the intake log with its reason.
5. References are checked across tables: a transaction of an item missing from the item list, or an expense booked to an unknown cost centre, is counted.

4. Preparation

Identical export lines are removed. Lines dated after the period end are left out. Refunds are kept as negative sales. Where a line has no cost, cost = quantity x the unit cost of the item list. A day on which a site that trades on at least 90% of days has no line is listed as a gap.

5. Sales and the drivers of change

Monthly sales, cost, gross profit, transactions, units and known customers are summed from the lines. The recent window is the last three complete months; it is compared with the same months a year earlier where they exist, otherwise with the three months before. For items sold in both windows, with q the units and p the average price: - volume = (Q1 - Q0) x sum(s0 x p0), where s0 is each item's share of units in the base window; - mix = sum((q1 - Q1 x s0) x p0); - price = sum(q1 x (p1 - p0)); - the sales of items sold in only one of the windows are shown on their own. Their sum equals the change in sales.

6. Profit by site

Over the last twelve complete months, by site: gross profit = sales - cost of sales; operating profit = gross profit - every expense booked to the site's cost centre. Expenses booked to no site are reported as head office and are not allocated. Sales per capacity unit, per square metre and per staff hour divide sales by the capacity, area and payroll hours of the site.

7. Items

Each item's sales, units, gross profit and margin over twelve months. An item is "popular" when its sales are at or above the median item, and its margin is "low" when it is below the larger of the agreed floor and 85% of the median margin. The four groups follow from these two cuts.

8. Customers

Using known customers only. New in a month: first purchase in that month. Lost: no purchase for the agreed number of days (90 by default), counted in the month the period runs out. Lost rate = lost / customers active in the period before. Comeback = share of new customers, old enough to tell, with a second purchase within 90 days. First-year value = average sales of a customer in the twelve months after the first purchase, and its gross profit at the gross margin of those purchases. Segments: Lost (beyond the lost period), New (first purchase within 60 days), At risk (frequent, half the lost period gone), Loyal (top quarter by number of visits), Occasional (the rest).

9. Price changes

For each item group, the median of its items' monthly average prices. A month-on-month change of at least the agreed size (5% by default) is a price event. Units per transaction in the three months after are compared with the

three months before; elasticity = that change / the price change. The lost rate of customers in the four months from the event is compared with the three months before.

10. Marketing

Over twelve months, by channel: spend, impressions, clicks and results from the platforms; new customers from the customer list, matched on the channel through which they joined. Return per unit = new customers x first-year gross profit per customer / spend.

11. Measures of the sector pack

The pack (`packs/<sector>.yaml`) names the measures of the sector as formulas of the standard totals (sales, cost, transactions, units, customers, capacity, area, staff hours, days, months). A measure whose inputs are missing is not shown.

12. Forecast

Log monthly sales are fitted by least squares on a linear trend, plus a dummy for each calendar month when there are 24 months or more. The error is measured by refitting without the last six months (a third of the series at most) and forecasting them: the mean absolute percentage error. The range is +/- 1.96 x the combined residual and test error, widening with the horizon.

13. What-ifs

New site: sales per capacity unit of the existing sites (median) x the capacity given; the chain's gross margin; the median running costs of a site other than rent and salaries; the rent and staff given. The first year ramps from half to full sales over the ramp months. Price change: units move by elasticity x change, unit cost unchanged; the elasticity is the one measured in section 9 where there is one, otherwise the agreed default.

13a. Purchasing and suppliers

Over the last 12 months: spend and share of spend by supplier; deliveries; lead time = date received – date ordered; late = received after the due date. Price change by supplier: average unit price of each item in the last three months against the first three, weighted by the supplier's spend on the items bought in both. For each item: the same comparison, the change in the average selling price over the same months (from the transactions), and the cost a year = change in unit price × units sold in 12 months; a rise above 5% that the selling price did not follow is counted as absorbed. Items bought from more than one supplier: best price = the lowest average price of a supplier; saving = (average paid – best price) × units bought.

13b. Stock

For each item and site over the 12 months: opening = the last count before the window; received and wasted = the sums reported with the counts; sold = units in the transactions (refunds negative); closing = the last count. Unexplained stock = opening + received – sold – wasted – closing, shown as a share of received; waste rate = wasted / received; days of cover = closing / (units sold in the last three months / days in them); stock value = closing × unit cost from the item list; excess value = units above 30 days of cover × unit cost; counts at zero = month-end counts with nothing on hand.

14. Assessment

Twenty-four health tests compare a measured value with thresholds agreed at kickoff (`tests:` in the configuration). A test whose data is missing is not run. Any FAIL gives "Needs attention", any WARN "Healthy, with actions", otherwise "Healthy". Each test that does not pass carries one action, an owner role, and a date 30 days (FAIL) or 60 days (WARN) after the extract.

15. Commentary

Findings are drafted from the numbers and reviewed by the analyst, who may add notes in `review/commentary_notes.yaml`.

16. Reproducibility

The run folder holds the configuration hash, input fingerprints, code version, every table as CSV and the log. The same inputs and configuration give the same tables.

17. Limitations

The limits of this work are set out in the report. No deviations from the standard methodology.